

1. If  $A = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 0 & -1 \\ 1 & 2 & 3 \end{bmatrix}$ , then show that  $A$  satisfies the equation  $A^3 - 4A^2 - 3A + 11I = 0$ , by using this equation find the inverse of  $A$ .

2.

If  $A = \begin{bmatrix} 1 & 1 & -1 \\ 2 & 0 & 3 \\ 3 & -1 & 2 \end{bmatrix}$ ,  $B = \begin{bmatrix} 1 & 3 \\ 0 & 2 \\ -1 & 4 \end{bmatrix}$  and  $C = \begin{bmatrix} 1 & 2 & 3 & -4 \\ 2 & 0 & -2 & 1 \end{bmatrix}$ ,

Find  $A(BC)$  &  $(AB)C$  hence prove that  $A(BC) = (AB)C$ .

3. The cost of 4 kg onion, 3 kg wheat and 2 kg rice is Rs 60. The cost of 2 kg onion, 4 kg wheat and 6 kg rice is Rs 90. The cost of 6 kg onion 2 kg wheat and 3 kg rice is Rs 70. Find cost of each item per kg by matrix method.

4. Find the area lying between the parabola  $y = 4x - x^2$  and the line  $y = x$ .

5. Calculate the volume of the solid bounded by the surface  $x = 0, y = 0, z = 0, x + y + z = 1$ .